

Overweightness and SES

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Overweightness and SES

1 Overweightness in the Netherlands: A Social Problem Hidden in Plain Sight

Walk through any Dutch city today and the numbers get to ignore. More than half of Dutch adults are overweight, one in six has obesity (Rijksinstituut voor Volksgezondheid en Milieu [RIVM], 2026), and if current trends continue, two in three people could be affected by 2050. What was once seen as a personal health issue has become a major social challenge. This became clear when Zorginstituut Nederland announced that bariatric surgery will now be covered by basic health insurance for teenagers aged 13 to 18 with severe obesity. Although only a small number of young people will qualify, the decision reflects growing recognition that lifestyle interventions alone are not always enough. To understand obesity, however, we must look beyond individual choices. In the Netherlands, obesity is strongly linked to social inequality. People with lower levels of education are significantly more likely to be overweight than university graduates (RIVM, 2026). Lower-income groups often have less access to healthy food, safe places to exercise, and supportive living environments. Research also shows that factors such as neighborhood safety, green spaces, and social cohesion influence obesity rates (OECD & European Observatory on Health Systems and Policies, 2025). The consequences extend beyond health. Obesity is associated with higher healthcare costs, lower earnings, increased sick leave, and reduced economic productivity. Children with obesity are more likely to face educational, social, and health disadvantages later in life. This argues that obesity in the Netherlands should be viewed as a social problem rather than simply a matter of personal responsibility. Addressing it effectively requires tackling the structural inequalities that make obesity more common among disadvantaged groups.

2 Data

2.1 Loading in data.

```
gewicht_regios <-  
  read_xlsx("data/Gezondheidsmonitor__regio__2024_04062026_114221.xlsx")  
saveRDS(gewicht_regios, "data/gewicht_regios.rds")  
  
overgewicht_over_tijd <-  
  read_xlsx("data/Lengte__onder__en_overgewicht_vanaf_1981_04062026_120636.xlsx")  
saveRDS(overgewicht_over_tijd, "data/overgewicht_over_tijd.rds")  
  
inkomen_dataset <-  
  read_xlsx("data/Inkomens_Huishouden.xlsx")  
saveRDS(inkomen_dataset, "data/inkomen_dataset.rds")  
  
opleiding_dataset <-  
  read_xlsx("data/Onderwijs_per_provincie.xlsx")  
saveRDS(opleiding_dataset, "data/opleiding_dataset.rds")  
  
gewicht_regios <-  
  readRDS("data/gewicht_regios.rds")  
overgewicht_over_tijd <-  
  readRDS("data/overgewicht_over_tijd.rds")  
inkomen_dataset <-  
  readRDS("data/inkomen_dataset.rds")  
opleiding_dataset <-  
  readRDS("data/opleiding_dataset.rds")
```

2.2 Explanation data

In this project, we utilize two datasets released by CBS (Statistics Netherlands) that collectively offer insight into the health of the Dutch population. The initial dataset, Gezondheidsmonitor; adults 18 years and older, region, 2024, centers on individuals aged 18 and above residing in the Netherlands. CBS compiled it in partnership with the National Institute for Public Health and the Environment (RIVM) and the local Public Health Services (GGDs). The data collection provides insights into health conditions, lifestyle choices, and social elements across various geographical tiers, such as municipalities, GGD regions, provinces, and at the national level. Essential indicators consist of self-assessed health, rates of chronic diseases, overweight and obesity, levels of physical exercise, involvement in sports, tobacco use, alcohol intake, and informal caregiving. The information is primarily shown as percentages of the adult demographic and offers a comprehensive regional overview of health and wellness in 2024. The second dataset, Lengte en gewicht van personen, ondergewicht en overgewicht; vanaf 1981, provides a longitudinal view of body weight and obesity trends in the Netherlands. This dataset includes individuals aged four and older and monitors trends from 1981 to 2025. It contains data regarding body mass index (BMI), moderate overweight, obesity, and underweight, categorized by age group, gender, and year. The data relies on self-reported weight and height, which CBS utilizes to determine BMI categories. In contrast to the initial dataset that highlights regional variations for one year, this dataset offers a national time series enabling researchers to explore changes in overweight and obesity across more than forty years.

Collectively, these datasets enhance each other effectively. The Health Monitor dataset offers an in-depth view of regional health disparities in 2024, whereas the height and weight dataset uncovers enduring national patterns of overweight and obesity. Integrating information from both sources allows for a clearer comprehension of obesity's prevalence in the Netherlands, its regional variations, and its historical trends.

These datasets are too big to show here, but the key variables will be explained now. In the first dataset, the health per region with 9 different values, only the region, lightly overweight and obesity columns are needed. In the rows not everything is important as well, but that will be discussed in the data cleaning. The second dataset, the percentage of overweight people over the years, is divided into underweight, overweight and obesity. Only overweight and obesity will be used, and of course the years. This will be used for total percentage of people, total percentage of men and total percentage of women. The third dataset contains information about income. The only necessary variables in this dataset are the province and the average income per person for these provinces. The last dataset is about education level for every province. This is divided into five different education levels that will be needed, and again the provinces. # 3 Quantifying

3 Data cleaning and creating new variables

```
overgewicht_1981 = overgewicht_over_tijd[1,5]
overgewicht_2025 = overgewicht_over_tijd[13,5]
#Selecting data for making the variable for the percentage change over time
```

```
proc_ver_overgewicht = ((overgewicht_2025 - overgewicht_1981) / overgewicht_1981) * 100
proc_ver_overgewicht[[1]]
```

```
## [1] 63.50365
```

To show the importance of this research a new variable is created, the percentage change of overweight people in the Netherlands is calculated over 44 years, which is 63.5 percent. This is obviously a huge increase which needs to be addressed and more attention given to.

3.1 Data cleaning final dataset one

These four different dataset needed to be merged into two dataset with only the necessary variables. Firstly, the overweight over time dataset was used. The data that was needed was the total percentage of total overweight and obesity, and the percentage of total overweight in men and women. Since these were not on the same rows, this was done by slicing and then full_joining them based on the year they were in. To this was a new variable introduced, the percentage point change compared to the last data point. Since this is impossible for the first year, NA got added there. This is to see how and how much the percentage of overweight people change, this gave the first final dataset looking like this (with two more columns).

```
## # A tibble: 13 x 6
##   Perioden Onder- en overgewicht-1 Onder- en overgewicht-2 Onder- en overgewicht-3
##   <dbl>                <dbl>                <dbl>                <dbl>
## 1 1981                27.4                4.4                30.2
## 2 1985                27.3                4.3                30.7
## 3 1990                30                5.6                33.8
## 4 1995                32.5                6.2                34.7
## 5 2000                37.7                8.1                40.3
## 6 2005                38.9                9.3                42.8
## 7 2010                41.7                9.8                46
## 8 2015                43                11.6               46.3
## 9 2020                44.4                12.1               47.2
## 10 2021                44.5                12.6               46.4
## 11 2023                44.2                13.6               46.4
## 12 2024                44.8                13.8               47.3
## 13 2025                44.8                13.8               46.3
## # i abbreviated names:
## # 1: 'Onder- en overgewicht, 4 jaar of ouder/Overgewicht (%)',
## # 2: 'Onder- en overgewicht, 4 jaar of ouder/Mate van overgewicht/Ernstig overgewicht (%)',
## # 3: 'Onder- en overgewicht, 4 jaar of ouder/Overgewicht (%)'.x'
## # i 2 more variables:
## # 'Onder- en overgewicht, 4 jaar of ouder/Overgewicht (%)'.y' <dbl>,
## # Percentage_difference <dbl>
```

From left to right the columns are; year, total percentage of overweight people, total percentage of obese people, total percentage of overweight men, total percentage of overweight women and percentage point change of the total percentage of overweight people.

3.2 Data cleaning final dataset two

Then the other dataset for different provinces was created. Firstly the dataset of education level was used. Two new variable got created, the total percentage of lower educated people, by adding two columns to eachother. Then the total percentage of higher educated people, also by adding two columns together. Both higher and lower education were split into two different columns. For middle level of education this was not the case. Then we sliced out all the regions that werent needed, so only the provinces were left. The next thing for this dataset that was taking extracting the needed columns from the income dataset. This was the only thing for this that was done. These two datasets were then added together so that their province name could be set to the specific name needed later on. Another dataset was needed to be added to this one. This dataset contained the percentage of overweight people per region, but was not divided by province but by GGD regions. These regions needed to be converted to provinces. These regions are always only in one province, making this fairly easy. The following function got used for this:

```
data_analysis <- function(regio1 = 0, data1 = 0, regio2 = 0, data2 = 0,
                          regio3 = 0, data3 = 0, regio4 = 0, data4 = 0,
                          regio5 = 0, data5 = 0){
  totaal_provincie <- regio1 + regio2 + regio3 + regio4
  data_provincie <- ((regio1 / totaal_provincie) * data1)
  + ((regio2 / totaal_provincie) * data2)
  + ((regio3 / totaal_provincie) * data3)
  + ((regio4 / totaal_provincie) * data4)
  + ((regio5 / totaal_provincie) * data5)

  return(data_provincie)
}
```

#The function above is created to calculate a new variable, the percentage of overweight people in the

The data that was needed for the populations of the regions was found on AlleCijfers (2026). Regio1, regio2, etc. were the amount of people living in the region, and data1, data2, etc was the percentage of overweight people. totaal_provincie was the total amount of people living in the province. Each factor was set to zero, because each province consists of a different amount of GGD regio's, and like this no errors would happen.

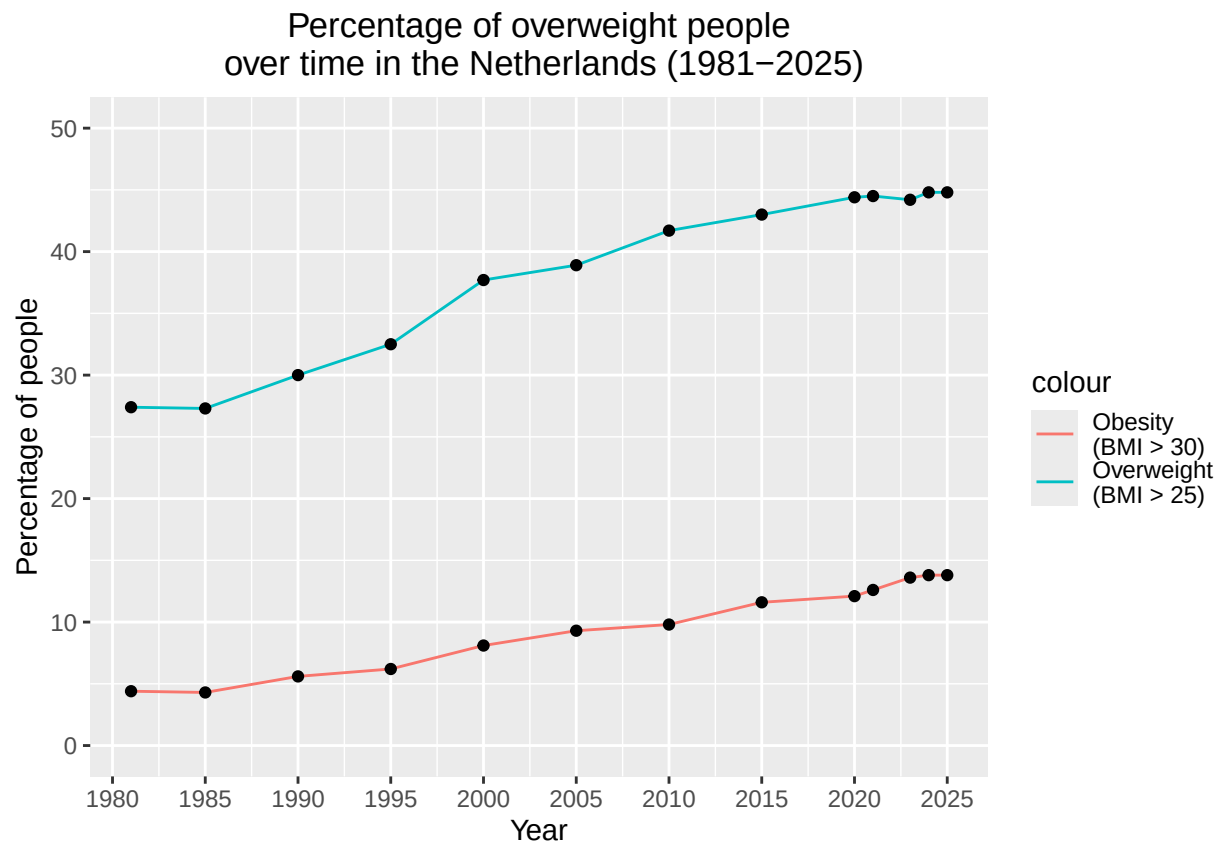
After all this data was created, and put into one data.frame, this data got left_joined by province to create the final dataset for the maps, looking like this, with 3 more columns not fitting:

```
## Joining with 'by = join_by(Regio)'
```

##	Regio	Overgewicht	Percentage_laagopgeleide
## 1	Groningen	49.70000	23.4
## 2	Fryslân	51.60000	26.6
## 3	Drenthe	55.90000	26.8
## 4	Overijssel	23.20834	26.1
## 5	Flevoland	56.00000	27.7
## 6	Gelderland	20.49976	26.0
## 7	Utrecht	44.10000	22.3
## 8	Noord-Holland	12.45783	23.7
## 9	Zuid-Holland	14.38822	27.2
## 10	Zeeland	51.60000	28.8
## 11	Noord-Brabant	14.49362	26.5
## 12	Limburg	25.00233	28.8

again, from left to right, including the last three columns that are not visible; Province, total percentage of overweight people, percentage of lower educated people, percentage of middle educated people, percentage of higher educated people, average income per person per province (x1000). These got then left_joined with the data for the map, to create one dataset to plot the 4 maps.

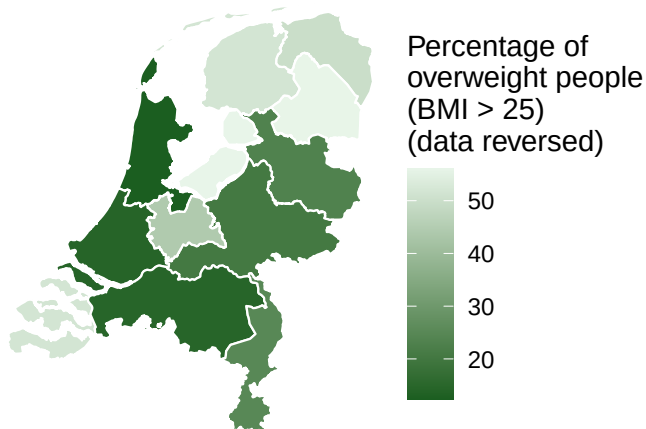
3.3 Overweight and obesity over time



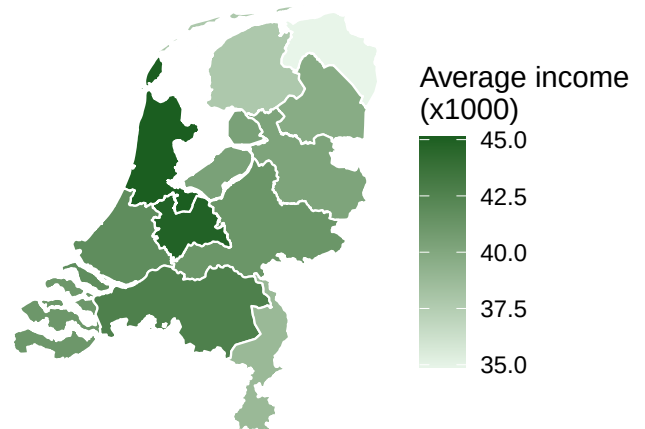
As shown in the graph, the percentage of overweight/obese people has increased at a relatively constant rate over time, with the exception of a larger increase between 1995 and 2000.

3.4 Maps of different provinces in the Netherlands for overweight, education and income

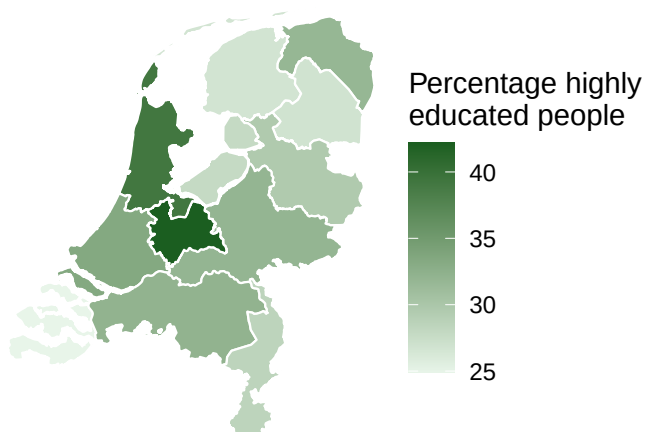
Percentage of overweight people by province in the Netherlands (2024)



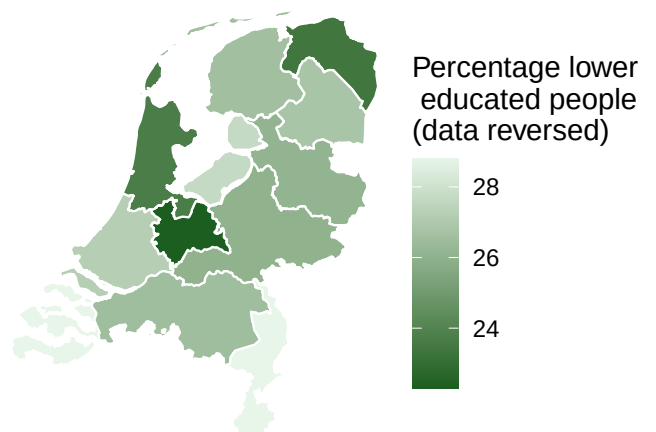
Average income by province in the Netherlands



Percentage of highly educated people by province in the Netherlands (2024)



Percentage of lower educated people by province in the Netherlands (2024)

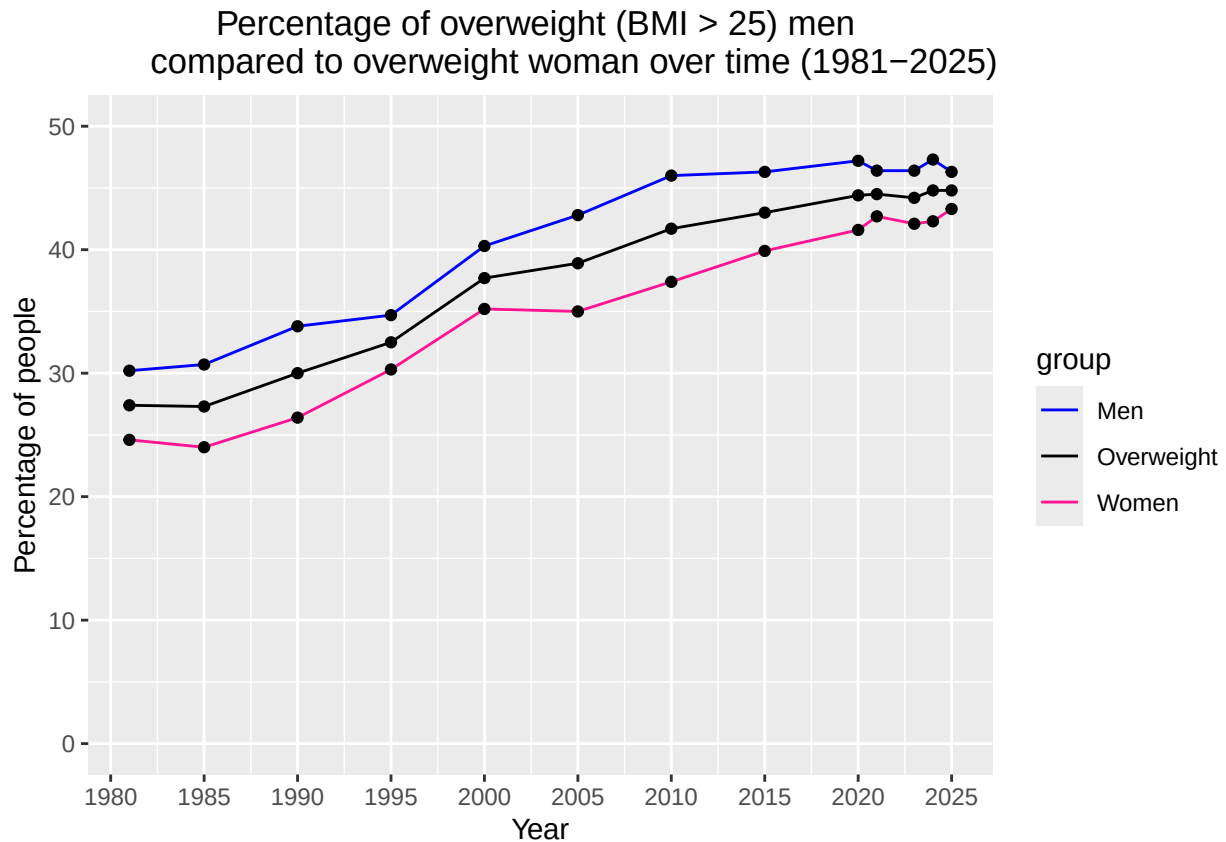


In the top left and down right, the data is reversed. This is done because this makes comparing the different maps easier and clearer.

As shown in the map representing the percentage of overweight and obese individuals, the provinces Utrecht, Noord-Holland, and Groningen appear to have the lowest prevalence rates. In contrast, Flevoland, Limburg and Drenthe have the highest percentage of overweight and obese residents. Furthermore a possible correlation can be found between overweight and obesity and educational level. The provinces with the lowest

prevalence rates also have the highest level of education, whereas Flevoland and Drenthe, which have a relatively lower education level, show a higher percentage of overweight and obese residents. The level of income and overweight appear to have a weaker relationship.

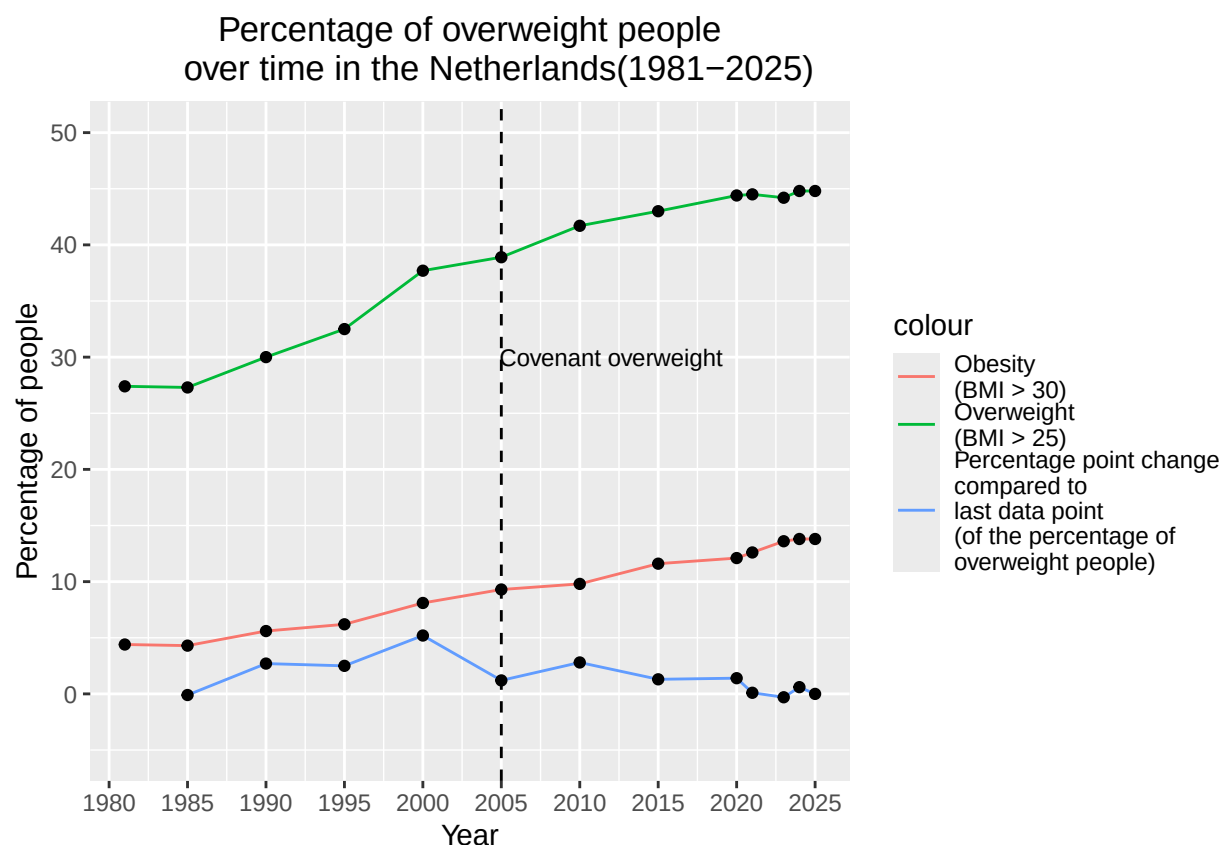
3.5 Overweight for men and women



Over the 44-year period shown, the percentage of overweight adults has always been higher among men than among women. Although this gap has shrunk in recent years, men still show a higher rates of overweight. furthermore, the percentage of overweight men and the percentage of overweight women have generally moved together over time.

3.6 Dutch covenant overweight

in 2005, the Dutch covenant overweight was introduced and signed by several organizations with the goal of preventing overweight and obesity. Here it is tested if the percentage of overweight and obese people has reduced since its introduction.



As shown in the temporal variation, the percentage of overweight/obese people has increased at a relatively constant rate over time. This rate seems to have slowed down a little from 2005 onwards in absolute terms, but this could be due to multiple reasons. Consequently, it is hard to determine a clear change following the introduction of the covenant on overweight and obesity, but an argument could be made that it did make a change.

4 - Discussion

The results of this study show that overweight and obesity have become increasingly important public health issues in the Netherlands. Between 1981 and 2025, the percentage of overweight individuals increased by approximately 63.5%, highlighting a strong upward trend over time.

Our regional analysis suggests a relationship between socioeconomic status and overweight prevalence. Provinces with a higher share of highly educated residents, such as Utrecht and Noord-Holland, generally have lower rates of overweight, while provinces with lower education levels, such as Drenthe and Flevoland, show higher rates. In contrast, the relationship between income and overweight appears less clear, suggesting that education may be more strongly associated with overweight than income.

We also found that men consistently have higher rates of overweight than women, although both groups follow a similar increasing trend over time. Furthermore, the introduction of the Dutch Covenant on Overweight in 2005 does not appear to have caused a clear reduction in overweight prevalence, although the growth rate may have slowed slightly.

A limitation of this study is that it relies mainly on descriptive analysis and visual comparisons, meaning that causal relationships cannot be established. Nevertheless, the findings indicate that overweight is linked to broader socioeconomic inequalities and should be viewed as more than just an individual lifestyle issue.

5 Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: https://github.com/davidhers/PfE_2.6_2026.git

5.2 How to reproduce

Overweight and SES project:

- README.md
- overweight_in_NL_2,6.Rmd
- data/
- gewicht_regios.rds
- opleiding_dataset.rds
- inkomen_dataset.rds
- overgewicht_over_tijd.rds

5.3 Reference list

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